

FU ZEYUAN

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github.com/Bardbar1an | bardbar1an.github.io/fuzeyuan-portfolio

EEE undergraduate at NTU (graduating Jun 2027) who builds LLM applications and AI agents end-to-end: stateful LLM agents (LangGraph, tool-using), QLoRA fine-tuning, LLM-as-a-judge evaluation, RAG, and on-device GGUF deployment. Available part-time (Tue–Sun) now; full-time from Dec 2026 through Jul 2027.

EDUCATION

Nanyang Technological University, Singapore — Aug 2023 – Jun 2027 (expected)

B.Eng. Electrical & Electronic Engineering

- **Relevant coursework:** Introduction to Data Science & AI (B+), Data Structures & Algorithms in Python (B+), Artificial Intelligence & Data Mining (in progress), Design & Innovation Project (A-), Engineering Mathematics I–II, Signals & Systems.

PROJECTS

AI Dungeon Master — locally fine-tuned LLM agent | May 2024 – Present | github.com/Bardbar1an/ai-dm

- **Agent design** — LangGraph stateful pipeline (rule-judge → narrator) with checkpointed memory: the LLM narrates while Python owns game state, dice and maps, grounding generation in deterministic game logic; Streamlit UI with Whisper voice input and TTS.
- **Fine-tuning** — QLoRA 4-bit fine-tune of Qwen2.5-7B-Instruct (Unsloth + TRL, rank-16) on 350 curated instruction dialogues distilled from API transcripts; trained on a consumer RTX 5080 (16 GB) with no cloud GPU spend.
- **Evaluation** — built an LLM-as-a-judge harness (weighted 1–5 rubric: immersion, player agency, rule fidelity) over 50 held-out prompts and ran head-to-head comparisons of cloud baseline vs. fine-tuned local model.
- **Deployment & RAG** — exported merged model to GGUF and benchmarked 4 quantization levels (llama.cpp): Q4_K_M cut size 69% (14.2→4.4 GiB) and lifted generation ~2.4× (5.6→13.6 tok/s); served via a self-written OpenAI-compatible FastAPI endpoint; RAG (ChromaDB + MiniLM) injects top-4 campaign-lore chunks into the narrator prompt.

FYP: Intelligent Control Design & Evaluation for a Human-Following Mobile Ground Robot | Aug 2026 – Present

- Model a ground robot and its environment for human-following scenarios; design a tracking controller combining Fuzzy Logic and Neural Networks.
- Implement and simulate the control system in Webots; evaluate tracking performance and robustness across scenarios.

MARS — Multi-Agent Pathfinding Simulator (EE3180 Design & Innovation Project) | Jan – May 2026 | grade A- | github.com/Bardbar1an/MAPF

- Built an interactive multi-agent pathfinding/relocation simulator (Python/Pygame): a centralized GlobalPlanner routes N agents to an exit/parking zone around obstacles, with smart-exit slot assignment and path smoothing.
- Modular control loop separating planner from simulation engine (time-stepped updates, live path rendering, start/reset/speed controls, pause-mode map editor); per-frame CSV logging for analysis.

ESP32-S3 On-Device Gesture Recognition | 2026

- Trained a 1D CNN (Keras) on self-collected MPU6050 IMU data and quantized float32 → int8 TensorFlow Lite Micro for on-device inference on an ESP32-S3 (≈95% float-accuracy target on 4 gesture classes; int8 within a few points).
- Streamed detected gestures to AWS IoT Core over MQTT with mutual-TLS certificates.

Intro to Data Science & AI module project | Aug – Oct 2024 | Team lead of 3 |

github.com/Bardbar1an/Cardiovascular-Diseases-prediction

- Led a team to build a scikit-learn pipeline (cleaning → training → evaluation) on the Kaggle cardiovascular disease dataset.
- Evaluated multiple classifiers and selected the tuned Decision Tree at ~73% test accuracy (GridSearchCV, 10-fold); coordinated the team to meet deadlines.

EXPERIENCE

Shinryo (Singapore) Pte Ltd | Electrical Intern | Jul – Dec 2025

- QA'd and standardized tender equipment data in Excel, automating repetitive checks with formulas and macros (XLOOKUP/COUNTIF) to cut manual errors.
- Supported tender submissions: cable-containment quantity take-offs and Interconnection Definition Documents (IDDs), coordinating with engineers and BIM teams on IDD standards.

National Service, Singapore Armed Forces | Electrical Technician | Feb 2021 – Jan 2023

- Installed, maintained and repaired signal systems (100% operational readiness for mission-critical operations), including emergency repairs during field exercises under time and resource pressure.

SKILLS

- **ML / LLM:** PyTorch, Hugging Face Transformers + TRL, Unsloth (QLoRA), LangGraph / LangChain, RAG (ChromaDB + sentence-transformers), LLM-as-a-judge evaluation, GGUF quantization (llama.cpp), TensorFlow Lite Micro, scikit-learn, Keras.
- **Programming:** Python, C++, TypeScript (basic), Linux shell, Git/GitHub, FastAPI, Streamlit, Pygame.
- **Hardware & simulation:** circuit analysis, soldering, electrical system maintenance, ESP32 / IMU, Webots.
- **Languages:** English (fluent), Mandarin Chinese (fluent).